

TRAVEL PLANNER AGENT

AI-Powered Personalized Travel Planning Assistant

Problem Statement No. 5

An intelligent AI-powered travel assistant that understands user preferences, budgets, and constraints to create personalized and optimized travel plans.

Project Proposal & Technical Design

Leveraging IBM Cloud Lite Services and IBM Granite

Project Domain: Artificial Intelligence / Cloud Computing
Core Technology: IBM Granite / IBM Cloud Lite
Application Type: AI-Powered Travel Assistant
Primary Goal: Personalized Travel Planning

1 Executive Summary

The **Travel Planner Agent** is an AI-powered intelligent assistant designed to simplify and personalize the travel planning process. Instead of requiring users to manually search across multiple platforms, the system acts as a centralized travel planning assistant.

The agent understands the user's:

- Destination preferences
- Travel dates and duration
- Budget constraints
- Transportation preferences
- Accommodation requirements
- Activities and sightseeing interests
- Weather and scheduling constraints

Using **IBM Granite** as the generative AI foundation and **IBM Cloud Lite services** for cloud-based deployment and integration, the system generates personalized itineraries and recommendations.

Core Value Proposition: The Travel Planner Agent transforms complex travel research into a single conversational experience that produces practical, personalized, and budget-aware travel plans.

2 Problem Statement

Planning a trip traditionally requires users to search through multiple sources for destinations, transportation, hotels, activities, weather information, and local recommendations.

This process can be:

- Time-consuming
- Difficult to personalize
- Expensive when planning is inefficient
- Challenging when schedules or weather conditions change
- Difficult to optimize within a fixed budget

The proposed Travel Planner Agent addresses these challenges by providing an intelligent conversational interface capable of understanding user requirements and generating a customized travel plan.

3 Project Objectives

The primary objectives of the system are:

1. Develop an AI-powered conversational travel assistant.
2. Understand natural-language travel requirements.
3. Generate personalized day-by-day itineraries.
4. Recommend transportation and accommodation options.
5. Consider user-defined budgets and constraints.
6. Incorporate weather and destination information.
7. Optimize schedules to reduce unnecessary travel.
8. Provide local activity and sightseeing recommendations.

9. Enable users to modify plans through conversation.
10. Deploy the application using IBM Cloud technologies.

4 Proposed Solution

The proposed system provides a conversational AI interface where users can describe their travel requirements naturally.

For example:

User Request:

“Plan a 5-day Kerala trip for two people within a budget of 20,000. I prefer nature, sightseeing, and comfortable accommodation.”

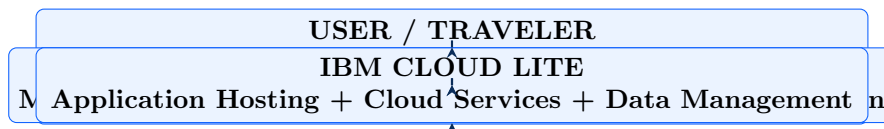
The Travel Planner Agent processes the request and generates:

- Suitable destinations
- Day-wise itinerary
- Estimated travel costs
- Transportation recommendations
- Accommodation suggestions
- Activities and sightseeing locations
- Weather considerations
- Budget summary

The user can subsequently ask the agent to modify the itinerary, change the budget, replace a hotel, add activities, or optimize the schedule.

5 System Architecture

The high-level architecture of the Travel Planner Agent consists of five major layers.



6 Core Features

6.1 1. Intelligent Travel Conversation

Users can communicate with the system using natural language. The agent interprets the user's intent, preferences, and constraints.

6.2 2. Personalized Destination Recommendation

The system recommends destinations based on:

- Budget
- Travel duration
- Interests
- Season
- Preferred activities

6.3 3. Automated Itinerary Generation

The agent generates a structured day-by-day itinerary containing destinations, activities, estimated travel time, and suggested schedule.

6.4 4. Budget-Aware Planning

The system considers the user's budget and provides an estimated breakdown for:

Category	Example Considerations
Transportation	Flights, trains, buses, local transport
Accommodation	Hotels and stays
Food	Estimated daily food expenditure
Activities	Entry fees and experiences
Miscellaneous	Local and emergency expenses

6.5 5. Weather Integration

Weather information can be used to improve activity planning and help users select appropriate travel dates and outdoor activities.

6.6 6. Route and Schedule Optimization

The agent can organize activities geographically and temporally to minimize unnecessary travel and improve the overall itinerary.

6.7 7. Dynamic Plan Modification

Users can interactively modify their plan.

Examples:

- “Reduce the budget to 15,000.”
- “Add one more day.”
- “Replace the hotel.”
- “Add more nature activities.”
- “Remove expensive activities.”

7 IBM Technology Stack

The project follows the mandatory requirement of using **IBM Cloud Lite services and/or IBM Granite**.

Technology	Purpose
IBM Granite	Generative AI and natural-language travel planning
IBM Cloud Lite	Cloud-based application infrastructure
Python	Backend logic and AI orchestration
Flask / FastAPI	Application programming interface
HTML / CSS / JavaScript	User interface
REST APIs	External travel-data integration
Weather API	Weather information
Maps / Location API	Routes and destination information
Database	User preferences and travel-plan storage

Mandatory Technology: IBM Granite and/or IBM Cloud Lite services will form the core IBM technology component of the proposed solution.

8 Role of IBM Granite

IBM Granite can serve as the intelligent reasoning and generation layer of the Travel Planner Agent.

Its responsibilities include:

1. Understanding natural-language travel requests.
2. Extracting user preferences and constraints.
3. Generating personalized travel recommendations.
4. Producing structured itineraries.
5. Summarizing travel information.
6. Responding to itinerary modification requests.
7. Maintaining conversational context.

A simplified processing flow is:

User Input → **Intent Extraction** → **Constraint Analysis** → **Travel Data** → **IBM Granite** → **Personalized Itinerary**

9 System Workflow

1. User enters travel requirements.
2. Application extracts relevant parameters.
3. User preferences and constraints are identified.
4. Required destination and travel information is collected.
5. IBM Granite processes the travel context.
6. The AI generates candidate travel plans.
7. Plans are evaluated against budget and constraints.
8. The optimized itinerary is presented to the user.
9. User can request modifications.
10. The system regenerates the plan based on the updated requirements.

10 Sample User Interaction

User:

Plan a 4-day trip to Kerala for two people with a budget of 20,000. I prefer nature, sightseeing, and comfortable stays.

Travel Planner Agent:

The system analyzes the budget, duration, preferences, and destination information and generates a personalized 4-day itinerary with:

- Day-wise destinations
- Accommodation suggestions
- Transportation options
- Activities
- Estimated expenses
- Weather considerations

11 Data and Personalization

The system can maintain structured information such as:

Data Category	Example Data
User Preferences	Nature, adventure, culture, shopping
Budget	Total and daily spending limits
Travel Dates	Start date and end date
Destination	Preferred city / region / country
Transportation	Flight, train, bus, car
Accommodation	Hotel category and location
Activities	Sightseeing and experiences
Trip History	Previous plans and preferences

This information allows the system to provide increasingly relevant recommendations during a travel-planning conversation.

12 Security and Reliability

The application should follow secure development practices, including:

- Secure handling of API credentials.
- Environment variables for sensitive configuration.
- Input validation and sanitization.
- Controlled access to stored user information.
- Secure API communication.
- Protection of personal travel preferences.

The system should also clearly distinguish between estimated information and confirmed real-time availability.

13 Expected Outcomes

The completed system is expected to provide:

- A functional AI-powered travel planning assistant.
- Conversational travel-plan generation.
- Personalized destination recommendations.
- Budget-conscious itinerary generation.
- Transportation and accommodation suggestions.
- Weather-aware activity planning.
- Interactive itinerary modification.
- Cloud-based deployment using IBM technologies.

14 Future Enhancements

Potential future improvements include:

- Real-time flight and hotel booking integration.
- Payment gateway integration.
- Voice-based travel assistance.

- Multilingual conversational support.
- Personalized recommendation learning.
- Emergency travel alerts.
- Offline itinerary access.
- Live traffic-aware route optimization.
- Calendar integration.
- Mobile application deployment.

15 Conclusion

The **Travel Planner Agent** provides an intelligent and personalized approach to travel planning by combining generative AI, cloud computing, travel information, and user preferences.

By leveraging **IBM Granite** and **IBM Cloud Lite services**, the proposed system can transform a user's natural language requirements into practical travel recommendations and optimized itineraries.

The project demonstrates the application of AI in a real-world domain while focusing on personalization, budget optimization, dynamic planning, and improved user experience.

Travel Smarter. Plan Better. Explore More.